

Berke Akkaya

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Research Profile

PhD in Quantitative Methods (2026) with a research background in machine learning, natural language processing, recommender systems, and empirical model evaluation. My work combines text representations with structured and interaction data to build and evaluate recommendation models, and examines how model performance varies across users and datasets. This includes a year of doctoral research in Germany (University of Passau) on LLM-based recommendation. I am interested in extending this work toward human/AI interaction and the societal effects of deployed AI systems.

Research Interests

Recommender systems; natural language processing and large language models; model evaluation and benchmarking; user modelling and heterogeneity in model performance; algorithmic bias and feedback loops; explainable AI; statistical modelling and experimental design.

Education

Istanbul University

Doctor of Philosophy (PhD), Quantitative Methods

Sep. 2021 – May 2026

Istanbul, Turkey

- Thesis: *Developing a Content-Aware Product Recommendation System Based on Natural Language Processing*. Combined sentence-transformer text representations with structured and interaction data for content-aware recommendation, evaluated using ranking metrics.

University of Passau

Doctoral Exchange Researcher, Chair of Data and Knowledge Engineering

Sep. 2024 – Sep. 2025

Passau, Germany

- Faculty of Computer Science and Mathematics, hosted by Prof. Dr.-Ing. Alsayed Algergawy. Research topic: enhancing a large language model based knowledge graph attention network for recommendation.

Istanbul University

Master of Science (MSc), Quantitative Methods

Sep. 2018 – June 2021

Istanbul, Turkey

- Thesis: *Analyzing User Interactions of an E-Commerce Website with a Pay-Per-Click Model*. BSc in Business Administration, Istanbul University, Sep. 2014 – June 2018.

Research Experience

Istanbul University, School of Business

Research Assistant, Department of Quantitative Methods

Nov. 2020 – Present

Istanbul, Turkey

- Design and run empirical studies on recommender systems and NLP: build end-to-end pipelines (data preparation, embedding-based representations, model training, evaluation) using Python, PyTorch, and RecBole across public and large-scale interaction datasets.
- Work on evaluation methodology: comparing model families under multiple, partly conflicting metrics, and studying heterogeneity in performance across users, including which user profiles are systematically harder to serve and what dataset and model properties predict this.
- Apply model-agnostic interpretability (SHAP) and explainability frameworks to decision-support settings, including transparency in algorithmic hiring; supervise undergraduate and graduate empirical projects on model evaluation and interpretability.
- Teach statistics, data mining, machine learning, and operations research; developed Python and R training material. Learning Coordinator, Institute of Business Economics (2021–2024).

Turing

Subject Matter Expert, Research Workflow Contributor

June 2026 – Present

Remote

- Contribute machine learning, recommender systems, and NLP expertise to a STEM taxonomy and research workflow project: specifying research-grade task decompositions and evaluation criteria for AI assistant outputs across applied research areas.

Istanbul Arel University

Research Assistant, Faculty of Economics and Administrative Sciences

June 2019 – Oct. 2020

Istanbul, Turkey

- Supported statistics teaching with a focus on model assessment and data preprocessing; served as Erasmus Institutional Coordinator.

Publications

Journal of Computer Science, 10(1), 53–91. Systematic survey of 67 studies across five methodological families, covering evaluation practice and open problems in the field.

With E. Celik. 25th International Business Congress, Ankara, Turkey, pp. 1–13 (full paper). Examines how automated and algorithmic steps in hiring are disclosed to candidates.

4th International Conference on Data Science and Applications, Istanbul, Turkey, pp. 142–152 (full paper). Empirical study of how a methodological choice interacts with dataset scale to change measured model performance.

With E. Sener & C. Gursu. 4th International Congress on Human-Computer Interaction, Optimization and Robotic Applications (HORA), IEEE, Ankara, Turkey, pp. 1–8 (full paper).

Izmir Journal of Economics, 37(3), 760–777.

Hacettepe Journal of Health Administration, 25(1), 17–28 (TR Dizin).

With N. Colakoglu. In *Dijitallesme Baglaminda Isletme Fonksiyonlarina Bakis*, Hiperyayin, Istanbul, pp. 303–327.

With N. Colakoglu. y-BIS Conference 2019: Recent Advances in Data Science and Business Analytics, Istanbul, Turkey, pp. 162–172 (full paper).

Technical Skills

Programming	Python, R, SQL
ML / NLP	PyTorch, TensorFlow, scikit-learn, SBERT / sentence-transformers, RecBole
Methods	Recommender systems, NLP and text representation, deep learning, classification, model evaluation and benchmarking, explainability (SHAP), experimental design, hypothesis testing
Statistical tools	statsmodels, SPSS, KNIME, Weka
Research workflow	Git, GitHub, reproducible evaluation pipelines, LaTeX

Academic Service, Memberships and Additional Experience

Service	Editor, <i>Alphanumeric Journal</i> (2025–present); scientific consultation for TEKNOFEST and IT Valley, Istanbul University (2024–present), advising student teams on applied machine learning projects
Memberships	Association for Computing Machinery (2024–present); International Society on MCDM (2022–present)
Funding	ISTQL, Sustainability of Cities and Brands, Technopark (2022–2023); ERASMUS+ scholarship (2024–2025)
Earlier roles	Membership Representative, Istanbul Foundation for Culture and Arts (2016–2018); Digital Banking Intern, Turkiye Is Bankasi (2017)
Languages	Turkish (native); English (C1, IELTS Academic); German (A2)
References	Prof. Dr. Seda Tolun Tayali, Istanbul University (stolun@istanbul.edu.tr); Prof. Dr.-Ing. Alsayed Algergawy, University of Passau (alsayed.algergawy@uni-passau.de); Prof. Dr. Nurdan Colakoglu, Istanbul Arel University (nurdancolakoglu@arel.edu.tr)